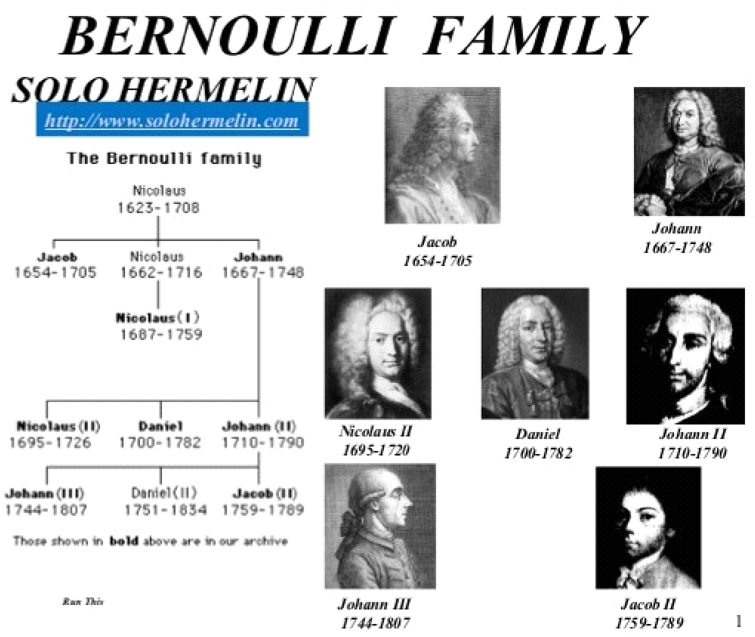
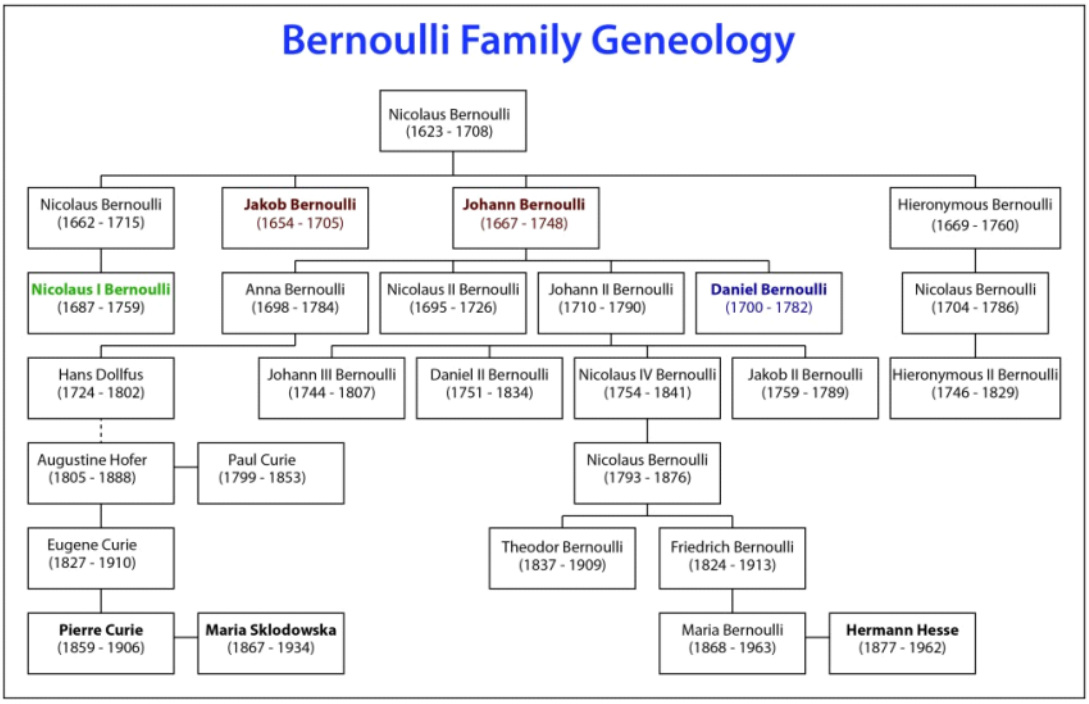
Lecture 13: Bernoulli's DE & Others

Thursday, 16 April 2026 9:56 am






LIST OF THINGS NAMED AFTER
MEMBERS OF THE BERNOULLI FAMILY
JOHANN BERNOULLI
JACOB BERNOULLI
JOHANN III BERNOULLI
JOHANN II BERNOULLI
NICOLAUS I BERNOULLI
NICOLAUS II BERNOULLI
DANIEL BERNOULLI
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I LOVE MATH
OH YEAH? NAME SEVEN MATHEMATICIANS
BERNOULLI
That's on me, I set the bar too low.
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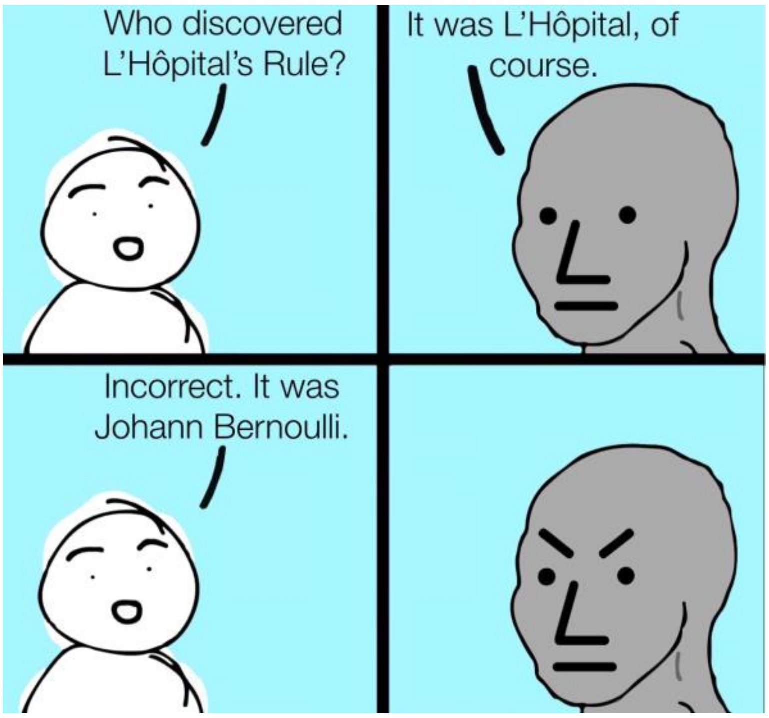


Bernoulli's Equation

$$y' + P(x)y = Q(x)y^n$$

If $y \neq 0$, then we have

$$y^{-n}y' + P(x)y^{-n+1} = Q(x)$$

put $w = y^{-n+1}$ then

$$w' = (-n+1)y^{-n}y'$$

and we have

$$\frac{1}{-n+1}w' + P(x)w = Q(x)$$

$$w' + (-n+1)P(x)w = (-n+1)Q(x)$$

which is a linear equation.

Example

Solve the differential equation

$$y' + xy = xy^2$$

Solution:

$$y = \frac{1}{1 + ce^{x^2/2}}$$

Example

* Solve the differential equation

$$5xy^2y' + y^3 = 32\,(1 + \ln x)\,y^{-2}, \quad x > 0\,,\, y \neq 0$$

Solution:

$$y^5 = 32 \ln x + \frac{c}{x}$$

Step 1: Multiply by y^2

$$5xy^4\frac{dy}{dx} + y^5 = 32(1+\ln x)$$

Step 2: Let $u = y^5 \Rightarrow \dfrac{du}{dx} = 5y^4\dfrac{dy}{dx}$

$$\Rightarrow x\frac{du}{dx} + u = 32(1+\ln x)$$

Step 3: Linear equation

$$\frac{du}{dx} + \frac{1}{x}u = \frac{32(1+\ln x)}{x}$$

$$\textbf{I.F.} = e^{\int \frac{1}{x}dx} = x$$

$$\frac{d}{dx}(xu) = 32(1+\ln x)$$

Step 4: Integrate

$$xu = 32\int(1+\ln x)\,dx = 32(x\ln x) + C$$

$$u = 32\ln x + \frac{C}{x}$$

Final Answer:

$$\boxed{y^5 = 32\ln x + \frac{C}{x}}$$

Solve the following IVP and find the interval of validity for the solution.

$$y' = 5y + \mathbf{e}^{-2\,x} y^{-2} \qquad y(0) = 2$$

$$y^2\, y' - 5y^3 = \mathbf{e}^{-2\,x}$$

$$v = y^3 \qquad v' = 3y^2 y'$$

$$\frac{1}{3}v' - 5v = \mathbf{e}^{-2\,x} \qquad \Rightarrow \qquad v' - 15v = 3\mathbf{e}^{-2\,x} \qquad \mu(x) = \mathbf{e}^{-15\,x}$$

$$v(x) = c\mathbf{e}^{15\,x} - \frac{3}{17}\mathbf{e}^{-2\,x}$$

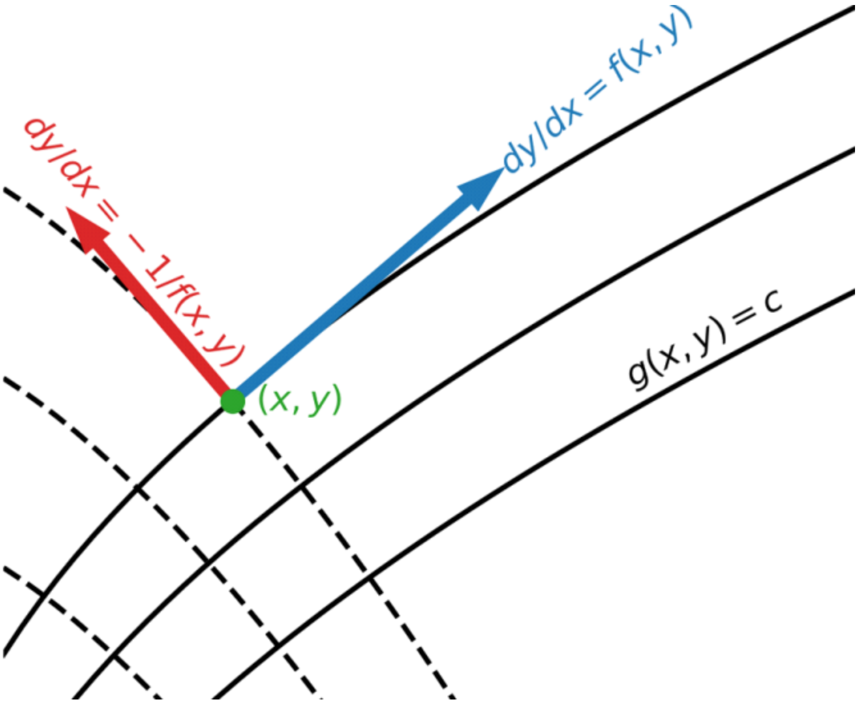
$$y^3 = c\mathbf{e}^{15\,x} - \frac{3}{17}\mathbf{e}^{-2\,x}$$

$$8 = c - \frac{3}{17} \qquad \Rightarrow \qquad c = \frac{139}{17}$$

$$y(x) = \left(\frac{139\mathbf{e}^{15\,x} - 3\mathbf{e}^{-2\,x}}{17}\right)^{\frac{1}{3}}$$


Orthogonal
trajectories


Orthogonal Trajectories
A family of curves perpendicular to another family of curves


dy/dx = f(x, y)
dy/dx = −1/f(x, y)
(x, y)
g(x, y) = c

Orthogonal Trajectories

1. Differentiate the family
$$F(x, y, c) = 0$$
implicitly to obtain an expression for $\frac{dy}{dx}$ (usually involving C).
2. Eliminate c to find the differential equation
$$\frac{dy}{dx} = f(x, y)$$
that describes the family.
3. The differential equation of the orthogonal trajectories is
$$\frac{dy}{dx} = \frac{-1}{f(x, y)}$$
4. Solve the resulting differential equation to obtain the orthogonal trajectories.

Example

Here we tackle a very simple example, consider the
origin:

$$x^2 + y^2 = c^2$$

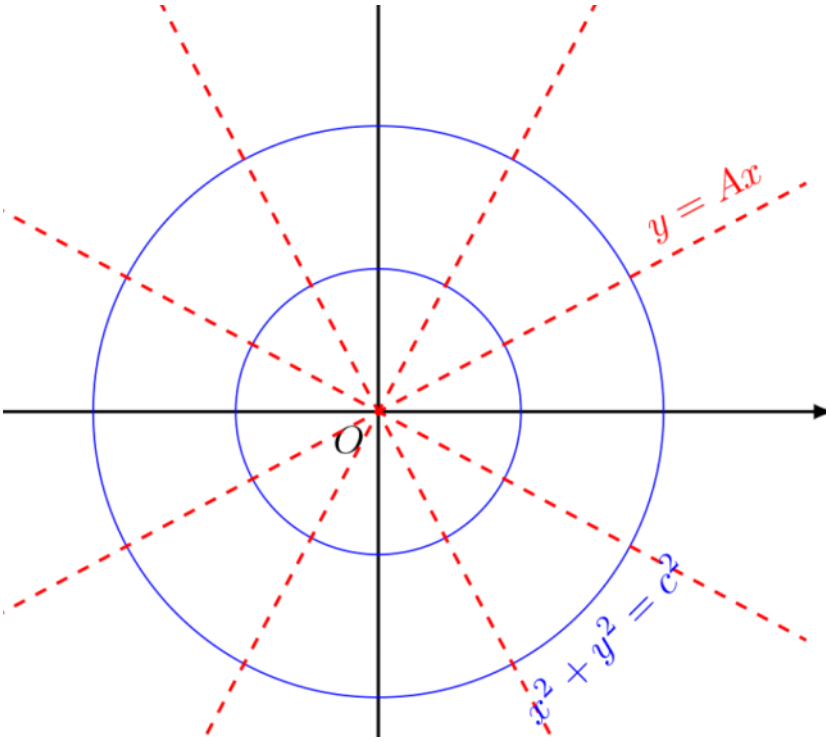
$$\begin{aligned} 2x + 2y\frac{dy}{dx} &= 0 \\ \frac{dy}{dx} &= -\frac{x}{y} \end{aligned}$$

$$\frac{dy}{dx} = \frac{y}{x}$$

$$\int \frac{dy}{y} = \int \frac{dx}{x}$$

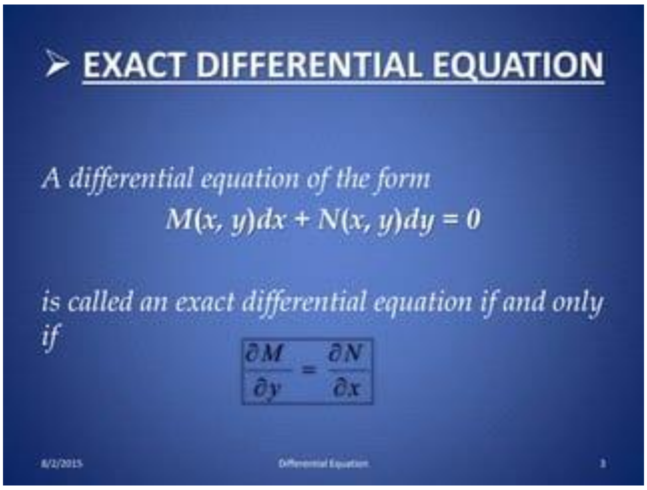
$$\ln y = \ln x + C$$

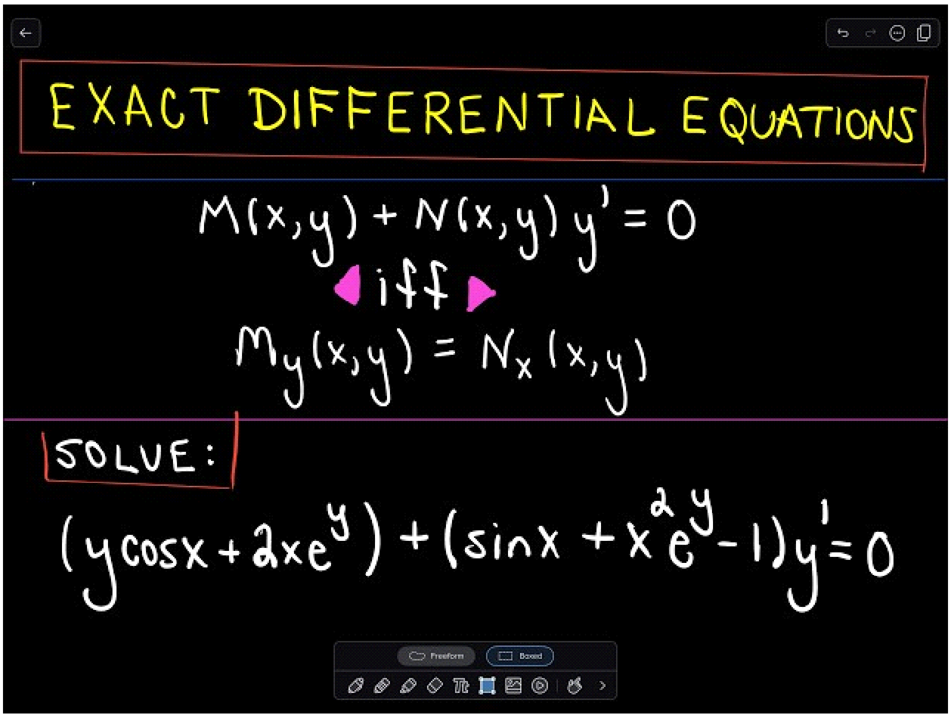
$$y = Ax$$



Ex. Find the orthogonal trajectories for $y = kx$.

Ex. Find the orthogonal trajectories for $x = ky^3$.





Exact Differential Equation

$$(2xy^2 - 4)dx + (2x^2y + 3)dy = 0$$

Non Exact
Differential Equations

$$M(x,y)dx + N(x,y)dy = 0$$

$$\mu(x) = e^{\int \frac{M_y - N_x}{N} dx}$$

$$\mu(y) = e^{\int \frac{M_y - N_x}{-M} dy}$$

Non - Exact Differential Equations (1)

Solve the differential equation

$$(x - y^2)dx + (2xy)dy = 0$$

M(x,y)dx + N(x,y)dy = 0

My ≠ Nx, for non-exact D.E's

SOLVING EXACT AND NON-EXACT DIFFERENTIAL EQUATION

1. $(x^2 + y^2)\,dx - xy\,dy = 0$
2. $(x + 4y^3)\,dy - y\,dx = 0$
3. $(4xy + y^2)\,dx - 2(x^2 - y)\,dy = 0$